The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals
NHS Foundation Trust

NEUROLOGICAL OBSERVATIONS

Since 1974, the Glasgow Coma Scale (GCS) has provided a practical method for bedside assessment of impairment of conscious level, the clinical hallmark of acute brain injury. The GCS provides an immediate prognostic guide and a baseline with which future examinations may be compared.

GLASGOW COMA SCALE (GCS) and SCORE

GCS looks at level of consciousness, therefore function of brain as a whole – non-invasive tool for measuring ICP

EYE OPENING		
SPONTANEOUSLY	4	Observed before you approach or speak to patient
TO SOUND	3	Call patient's name
TO PRESSURE	2	Apply pressure to side of finger *Central stimulus to supraorbital nerve will cause grimacing and eye closure*
NONE	1	Ensure stimulus adequate

Eye opening looks at arousal mechanisms and control of the eyes in the brainstem

Spontaneous opening – in absence of stimulation – is highest response that can be recorded but should not be equated to "alertness" or "awareness"

Even when brain damage is severe, patients who survive will eventually open their eyes, usually within 2 to 4 weeks

VERBAL RESPONSE		
ORIENTATED	5	Knows person, place, time *Should know who they are, where and why they are there and know month, year, season - if answers one or more component wrongly then record as confused*
CONFUSED	4	Talking in sentences but disorientated to time and place *Patient attends to questioning but shows disorientation*
WORDS	3	Utters occasional words rather than sentences *Often abusive words elicited by noxious stimuli rather than spontaneous*
SOUNDS	2	Groans or grunts
NONE	1	No verbal response *If patient has ET tube or tracheostomy record 'T' here*

Verbal response assesses comprehension and transmission of sensory input & ability to articulate a reply

An orientated response requires a defined minimum in three fields of information:

- Person – their name
- Place – their location (in hospital or other specific place)
- Time – the month

Dysphasia reflects focal damage to one of speech centres – but may be difficult to distinguish between confusion and disturbances in language function. The dysphasic patient usually attempts to guess questions or commands or show some indication of trying to co-operate with questioning even if unable to respond correctly. Expressive dysphasia more common and more obvious than receptive dysphasia

Expressive dysphasia – spontaneous, non-fluent speech but comprehension normal

Receptive dysphasia – spontaneous, fluent speech but comprehension impaired

MOTOR RESPONSE		
OBEYS COMMANDS	6	Give patient a simple command such as Hold up your arms' 'Stick out your tongue' 'Squeeze my hand' *Beware of reflex hand grasp in unconscious patient*
LOCALISES	5	Apply pressure to supraorbital nerve *Patient must bring hand to chin or above – if trauma to eyes use trapezius pinch*
FLEXES	4	Elbow bending recorded as flexing
ABNORMAL FLEXION	3	Elbow bending accompanied by spastic flexion of wrist
EXTENSION	2	Straightening of elbow recorded as extending
NONE	1	Ensure stimulus adequate

Motor response is the most important aspect of GCS

The best response is recorded as an indication of the functional state of the brain as a whole – what the brain is still capable of

Only the response of the upper limbs is recorded as these are more reliable than lower limb responses that could be purely spinal reflexes

if the patient does not obey commands, motor activity has to be assessed by applying a pressure stimulus (pressure on supraorbital nerve) – if patient has eye swelling or orbital fractures apply trapezius pinch instead

"Localising" implies a connection between location of sensory input and specific movement made in response; recommended standard is that hand is brought above the clavicle towards a stimulus on the head or neck - Bringing hand to opposite side of body not sufficient

if no response to central stimulus then a peripheral stimulus should be applied

ASSESSMENT of LIMB POWER *(focal sign)*

MRC GRADING SYSTEM

GRADE 5	Full power	*Assessing limb power enables comparison of one side of brain with the other identifying which side is most affected by the injury*
GRADE 4	Movement against resistance	
GRADE 3	Movement against gravity	
GRADE 2	Movement	
GRADE 1	Flicker	
GRADE 0	None	

PUPILLARY RESPONSE *(focal sign)*

CHECK DIRECT AND CONSENSUAL RESPONSE

Size	*Pupillary response is a critical component of the neurological examination and is the only clinical sign of raised ICP once the patient is sedated/ intubated / ventilated – but very late sign!*
Shape	
Equality	
Reaction to light	

VITAL SIGNS

Non-neurological complications, dysfunction or failure, are common after brain injury

HR	Monitor changes in rate / rhythm
BP	Avoid hypo / hypertension - *BP parameters should be set in acute injury*
RR	Monitor for hypo / hyperventilation - *GCS ≤8 requires intubation - consider also for rapidly deteriorating GCS or seizures especially when sedative drugs required to control (benzodiazepines)*
TEMP	Hyperthermia in acute brain injury worsens outcome – *investigate & treat*
Blood Glucose	Hypo / Hyperglycaemia should be avoided in acute brain injury

PATIENT CHANGES REQUIRING IMMEDIATE MEDICAL REVIEW

- Development of agitation or abnormal behaviour
- A sustained (for at least 30 minutes) drop of one point in GCS level (greater weight should be given to a drop of one point in the motor score of the GCS)
- Any drop of two points or more in GCS level regardless of duration or sub-scale
- Development of severe headache or persistent vomiting
- New or evolving neurological symptoms or signs (such as unequal pupils, limb weakness)